

# Integrated approach to the structural assessment of the Rai tower of Caltanissetta

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**Abstract.** The structural assessment of unconventional existing structures is often constrained by the limited availability of reliable design documentation, the difficulty of conducting extensive destructive testing, and the absence of specific standardized procedures. Under these conditions, defining an investigation protocol cannot rely on the simple application of standard schemes; instead, it requires an integrated approach combining local observations, global measurements, monitoring, and numerical modeling. This paper describes the methodological framework adopted for the RAI tower in Caltanissetta—a tall, guyed steel structure characterized by high slenderness, a spherical base hinge, and a significant dependence on the tension state of the guy cables. Key experimental activities are outlined, focusing on indirect mechanical characterization, the hole-drilling technique, guy cable tension verification, operational dynamic identification, and the monitoring of tension and environmental conditions. The study highlights the role of cross-validating data in reducing epistemic uncertainties and proposes an operational workflow applicable to other unconventional, data-scarce steel structures..

**Keywords:** Guyed towers, metal structures, unconventional structures, investigation protocol, Hole Drilling, Operational Modal Analysis, structural monitoring.

## 1 Introduction

The assessment of existing structures is typically based on a progressive investigative process, in which surveying, material testing, analysis of construction details, and numerical modeling combine to define the knowledge level and confidence factors. For standard buildings and many common infrastructure types, designers have access to regulatory standards and technical documents outlining the criteria, minimum objectives, and operational procedures for establishing an investigation plan.

The situation changes significantly when the subject of the assessment is an unconventional metal structure—characterized by large-scale geometry, a unique structural scheme, strong interaction with environmental loads, and design documentation that cannot be fully verified. In such cases, the very concept of "knowledge level" requires reinterpretation: simply increasing the number of tests is insufficient; instead, a coher-

ent information system must be constructed—one capable of linking local measurements, global response, and structural model parameters.

International literature on guyed towers highlights, in particular, the complex dynamic behavior of these structures and the high sensitivity of their response to guy pretension, boundary conditions, non-linear cable modeling, and wind loads [6–10]. In this context, experimental measurements play a crucial role not only in describing the structure's condition but also in calibrating and validating the numerical model used for subsequent assessments.

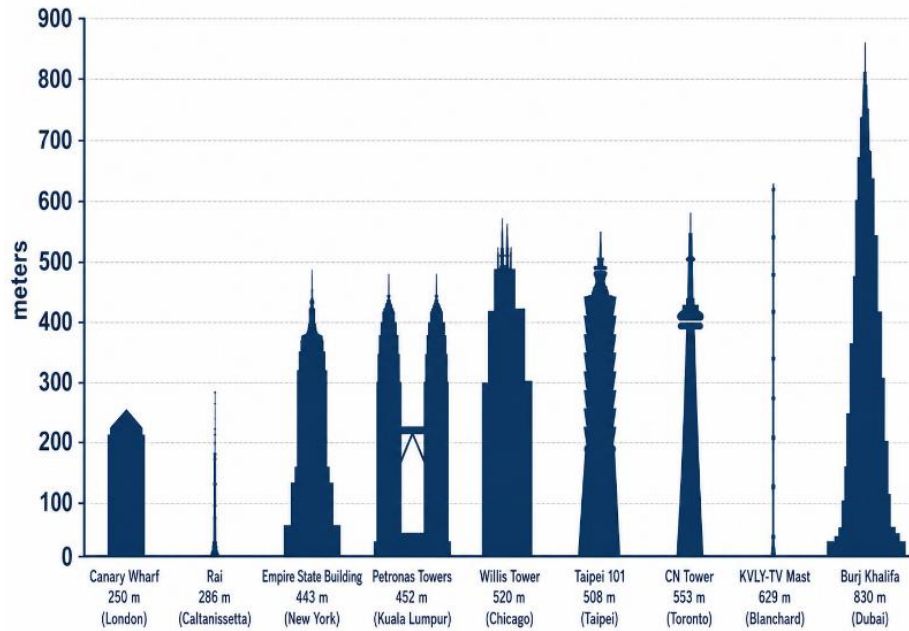
The RAI tower in Caltanissetta serves as a prime example of these challenges. It is a very tall steel lattice tower, stabilized by a system of guy wires and anchored at the base via a spherical hinge. This configuration makes the structure's behavior highly dependent on the tension state of the guy wires, boundary conditions, and environmental loads—particularly wind. This paper does not aim to present a final safety assessment of the structure—conducted by other parties based on available data—but rather to illustrate the rationale behind the design and implementation of an integrated investigation protocol.

## **2 Introduction**

### **2.1 General description of the work**

The RAI transmitting station at Caltanissetta-Sant'Anna, built between 1949 and 1951, served for decades as one of the region's key radio broadcasting infrastructures. Standing approximately 286 meters tall, the tower consisted of a metal lattice mast—comprising primarily angle-section members—stabilized by multiple tiers of radially arranged guy wires and connected to the foundation via a base assembly functioning much like a spherical hinge.

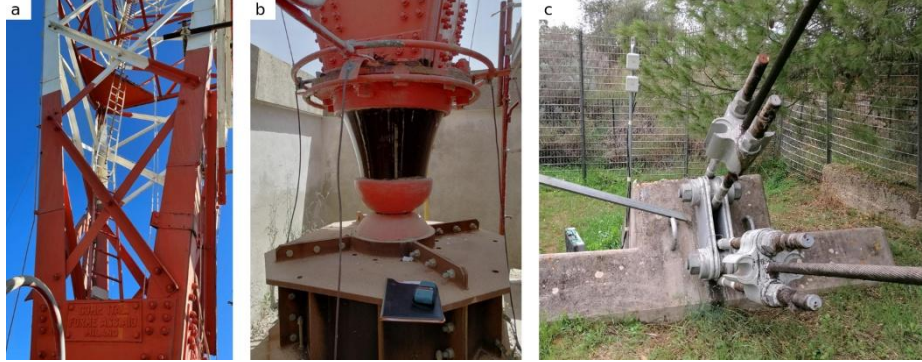
Given its height, the structure was a landmark feature of the Italian landscape; in terms of scale and visual impact, it rivaled some of the world's most famous vertical structures, even though it belonged to a different construction category than skyscrapers or freestanding towers.



**Fig. 1** Size comparison between the RAI tower in Caltanissetta and some major international vertical structures.



**Fig. 2** General view of the RAI tower in Caltanissetta, a guyed metal structure.



**Fig. 3** Characteristic features of the tower: lattice shaft made of angle sections, spherical base hinge, and stay anchorages.

The presence of the base hinge and the stay cables gives the structure a behavior fundamentally different from that of a freestanding tower. The metal mast cannot be analyzed in isolation: overall equilibrium depends on the distribution of tensile forces in the stays, their geometry, the stiffness of the anchorages, and the dynamic response of the entire system. An uncontrolled change in the tension of one or more stays can alter the equilibrium configuration and, under adverse conditions, compromise the tower's overall stability.

## 2.2 Operational and knowledge-related challenges

The design of the investigation protocol was shaped by two main challenges. The first concerns the lack of specific procedural standards for structures of this type. While regulatory guidelines or supporting documents exist for existing buildings and infrastructure such as bridges and tunnels, a tall, guyed metal tower requires specific choices tailored to the structural function of individual components and the risks associated with testing operations.

The second challenge involves the impossibility of adopting an investigation plan based on extensive metal sampling. Removing significant portions of the elements could have compromised local load-bearing capacity or introduced further uncertainties. Furthermore, the absence of laboratory samples made it impossible to calibrate non-destructive tests in the standard manner. The assessment strategy therefore had to rely on indirect or semi-invasive techniques, targeted repetition of measurements, and cross-validation of independent results.

## 3 Definition of the integrated protocol

The general logic of the protocol was to replace the traditional sampling–laboratory testing–result extrapolation scheme with an integrated, redundant process. Each investigation technique was selected not only for the specific data it provided but also for its contribution to verifying the consistency of the overall picture.

Local data provided information on the actual condition of the elements; global data allowed for the interpretation of the structure's response as a whole; and the numerical model served as a synthesis tool to verify the compatibility between hypotheses, measurements, and observed behavior.

Table 1. Summary of the integrated investigation protocol.

**Table 1** Captions should be placed above the tables.

| Scope of the investigation          | Learning objective                                                        | Main output                                           |
|-------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------|
| Microhardness and local inspections | Indirect estimation of mechanical properties                              | Classification and comparison of homogeneous elements |
| Hole Drilling                       | Assessment of the local in-situ stress state                              | Principal stresses and orientation                    |
| Stay cables and anchorages          | Determination of effective pull and monitoring of the unloading procedure | Stay loads and released deformations                  |
| OMA dynamic identification          | Overall assessment of dynamic behavior                                    | Natural frequencies and principal mode shapes         |
| Tension and wind monitoring         | Correlation between environmental actions and variations in stress        | Time series of deformation, strain, and wind          |
| FEM modeling                        | Interpretative synthesis of experimental data                             | Model calibrable against independent observables      |

## 4 Local investigations: materials and stress state

### 4.1 Mechanical characterization via indirect tests

The mechanical characterization of the metal elements was carried out using microhardness testing—a non-destructive method chosen to assess the variability of the material's properties. As samples could not be extracted for direct mechanical testing, these measurements provided localized, comparable data across different elements, aiding in the definition of homogeneous groups and the selection of parameters for the analytical model.

The value of this test lies not merely in estimating a mechanical property, but in the ability to extend the assessment to multiple points on the structure without causing significant damage. For tall, existing metal structures, this approach increases the density of data gathered during the survey without compromising the local structural integrity of the elements being examined.

### 4.2 Principles of the hole-drilling technique

The assessment of the in-situ stress state was conducted using the hole-drilling technique. The test principle relies on local stress relief: the controlled removal of a small volume of material via micro-milling causes a redistribution of pre-existing stresses,

generating elastic strains that can be measured on the surface using a strain gauge rosette applied around the drilling point.

Assuming a plane stress state and linearly elastic material behavior, the released strains measured along the rosette axes can be related to the principal stress components using calibration coefficients that account for the hole geometry, drilling depth, and the material's elastic properties. In summary, by using the released strains  $\epsilon_a$ ,  $\epsilon_b$ , and  $\epsilon_c$  and the calibration coefficients A and B, the procedure allows for the determination of the principal stresses  $\sigma_1$  and  $\sigma_2$  and the angle  $\theta$  of the principal directions..

$$p = \frac{\epsilon_a + \epsilon_c}{4A} \quad (1)$$

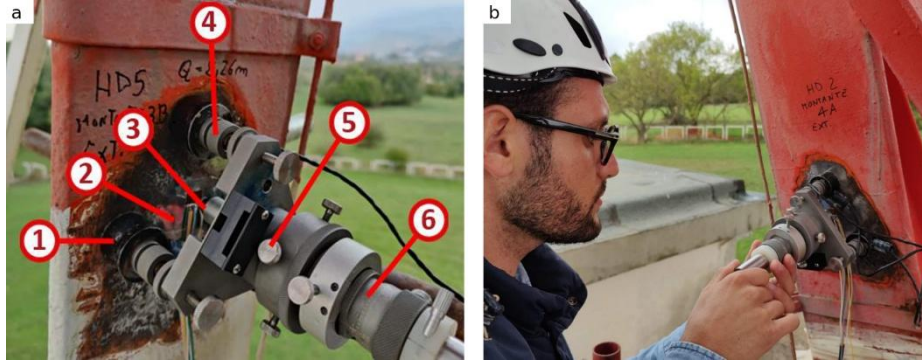
$$q = \frac{\epsilon_c - \epsilon_a}{4B} \quad (2)$$

$$t = \frac{\epsilon_a + \epsilon_c - 2\epsilon_b}{4B} \quad (3)$$

$$\sigma_{1,2} = p \pm \sqrt{q^2 + t^2} \quad (4)$$

$$\tan 2\theta = \frac{t}{q} \quad (5)$$

Principal stresses and principal stress directions are particularly useful in lattice structures, where the direction of actual stress can be influenced by eccentricities, connections, local constraints, and redistributions that are not immediately deducible from geometry alone. Compared to more invasive stress-relief methods, this technique allows for a local measurement of the stress state while limiting material removal to a very small volume.



**Fig. 4** Execution of the hole-drilling test and details of the measuring device: (1) magnetic bases for attachment to the metal structure; (2) strain gauge rosette; (3) micro-milling cutter; (4) adjustable feet for aligning the device with the measurement surface; (5) device for micrometric adjustment of the drilling point; (6) slider for micrometric adjustment of the milling cutter's removal depth.

A summary of representative results obtained using the hole-drilling method is presented in Table 2. The values confirm the utility of the test in determining not only the

magnitude of the local stress state but also the variability across different elements and elevations of the structure.

**Table 2** Representative results of Hole Drilling tests.

| <b>Trial</b> | <b>Elevation</b>  | <b><math>\theta</math> [°]</b> | <b><math>\sigma_1</math> [MPa]</b> | <b><math>\sigma_2</math> [MPa]</b> | <b>Summary reading</b>        |
|--------------|-------------------|--------------------------------|------------------------------------|------------------------------------|-------------------------------|
| HD1          | Elevation 0,00 m  | 43,94                          | 1,73                               | -2,60                              | Limited local stress state    |
| HD2          | Elevation 0,00 m  | 10,63                          | -4,11                              | -10,30                             | Predominant local compression |
| HD3          | Elevation 0,00 m  | -38,11                         | -23,71                             | -32,46                             | More pronounced compression   |
| HD4          | Elevation 12,10 m | -21,44                         | -3,93                              | -2,20                              | Limited stress state          |
| HD5          | Elevation 8,26 m  | 20,38                          | 38,93                              | 32,79                              | Higher local dose             |

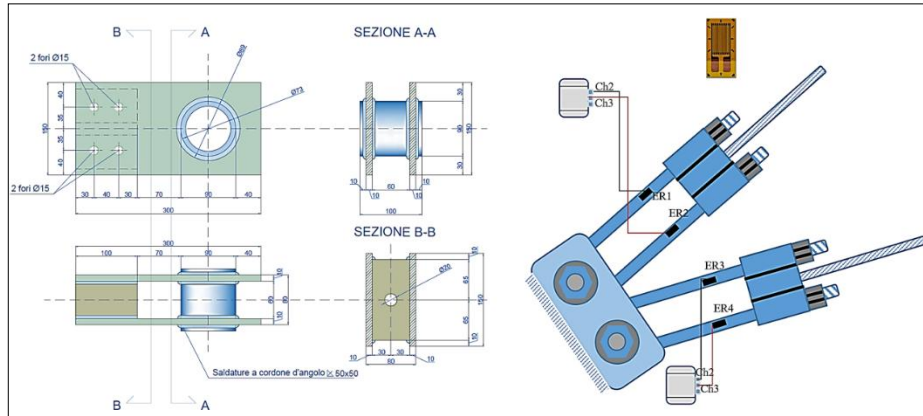
## 5 Verification of stay tensile load

Determining the tensile load in the stay cables was one of the most critical tasks in the protocol. In a stayed tower, cable tension is not a secondary detail but a parameter directly linked to lateral stiffness, natural frequencies, stress distribution, and the structure's overall equilibrium. Experimental determination was therefore essential both for understanding structural behavior and for setting up the numerical model.

The procedure involved instrumenting the anchorage forks with electrical strain gauges. Subsequently, the load was progressively transferred from the fork to a parallel system consisting of hydraulic jacks arranged symmetrically around the stay cable. During this transfer, the strain release in the fork was continuously monitored; once the released strain began to stabilize, the fork was considered fully unloaded, and the load reading from the parallel system was taken as an independent measure of the pre-existing tension.

This approach served a dual purpose: on one hand, it allowed for the estimation of the load originally borne by the fork by comparing the released strain with the load recorded by the jacks; on the other, it acted as an operational safety indicator. The test was designed not to increase the tension in the stay cable, but merely to take over the existing load. An excessive increase in load could have altered the tower's equilibrium state—a particularly sensitive factor given the presence of the spherical hinge at the base.





**Fig. 5** Verification of stay tension: load transfer scheme to a parallel system.



**Fig. 6** Verification of stay tension: anchoring system and instrumentation.

In summary, the operational sequence involved: installation and verification of strain gauges on the forks; gradual activation of the parallel jacks; monitoring of strain release; halting the procedure upon signal stabilization; and a final comparison between the released strains and the load recorded by the jacks. Selected representative values derived from the test records are summarized in Table 3.

| Stay group | Stay | Indicative load | Pressure [bar] | Stopping criterion                   |
|------------|------|-----------------|----------------|--------------------------------------|
| upper      | 1-4  | 150,0-159,0     | 437,5-465,0    | Stabilization of deformation release |
| lower      | 1-4  | 168,0-171,0     | 490,0-500,0    | Stabilization of deformation release |

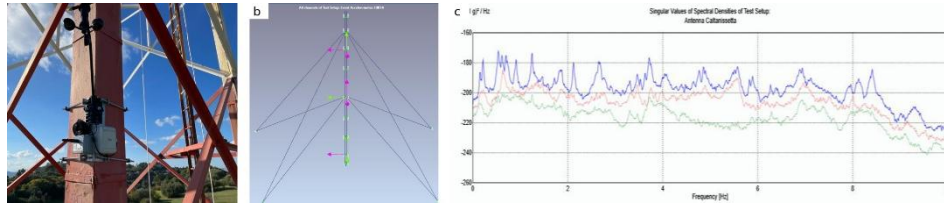


## 6 Dynamic identification and monitoring

### 6.1 Operational Modal Analysis

Local inspections and stay-cable tension checks provide essential information on individual components but are insufficient on their own to describe the tower's global behavior. Consequently, the protocol included an in-service dynamic identification campaign using Operational Modal Analysis, aimed at estimating the structure's natural frequencies and primary mode shapes.

The tower's considerable height and access difficulties necessitated a streamlined experimental setup. Three triaxial accelerometers were installed at different elevations along the shaft to sample the structure's dynamic response across its height. Spectral analysis was performed on both the overall three-dimensional system and the subsystems associated with the principal directions, utilizing singular values of the spectral density to identify dominant modes. From a methodological standpoint, the most significant result is that a limited number of sensors can yield useful dynamic data when their placement is designed based on expected mode shapes and actual operational constraints. For slender, very tall structures, this approach allows for the acquisition of global information without requiring extensive measurement campaigns, which are often unfeasible due to safety concerns or prohibitive costs.



**Fig. 7** Dynamic identification phases: accelerometer installation, setup representation, and spectral analysis using singular values of the spectral density.

### 6.2 Tension and wind monitoring

Alongside dynamic measurements, the protocol included continuous monitoring of the strain state of the stay-cable anchorages and wind conditions. The objective was to correlate stress variations observed in the anchorage elements with environmental loads, distinguishing between the system's routine oscillations and variations associated with gusts or specific wind conditions.

This integration is significant for stayed towers, where structural behavior cannot be decoupled from environmental conditions. Simultaneous measurement of strain and wind allows stress peaks to be interpreted not as isolated values, but as the system's response to a specific external load. The monitoring data thus become valuable both for assessing in-service behavior and for calibrating the assumptions made in the numerical model.

## 7 Role of experimental data in FEM modeling

The numerical modeling of a cable-stayed tower with limited available data is characterized by a large number of uncertain parameters. Key among these are material mechanical properties, the stiffness of truss elements, base boundary conditions, actual geometry, cable tension, and the representation of anchorages. In the absence of comprehensive documentation, these parameters cannot simply be adopted from design drawings; instead, they must be verified and updated based on experimental evidence. Microhardness tests contribute to defining mechanical parameters; the hole-drilling method provides insights into the local stress state; cable inspection allows for the input of tension values consistent with actual conditions; and dynamic identification offers global observables—such as frequencies and mode shapes—against which the model's consistency can be evaluated. The comparison between measurements and the model can thus be viewed as an iterative process in which a preliminary model is progressively updated by incorporating available experimental data.

In this context, the modeling objective is not to produce a representation that is formally detailed yet unvalidated, but rather to construct an analytical model consistent with the available observables. This approach helps reduce the epistemic uncertainty associated with an unconventional structure by making explicit which experimental data support the adopted assumptions and which require further investigation.

## 8 Discussion

The case of the RAI tower in Caltanissetta highlights how, for unconventional metal structures, defining the investigation protocol cannot be viewed merely as a preliminary step to the subsequent structural assessment; rather, it must be treated as a genuine design activity. Unlike standard buildings or common infrastructure, the investigative approach must be tailored to the specific characteristics of the load-bearing system, accessibility conditions, and the risks associated with individual testing operations.

The choice of investigation techniques was driven by the need to gather meaningful data without disrupting the structure's equilibrium or resorting to extensive destructive sampling of key elements. The protocol's value lies in the ability to compare diverse types of data: indirect mechanical properties, local stress states, guy-wire tension, global dynamic response, and wind-induced variations. Consistency among these parameters is more significant than the sheer volume of tests, particularly when the structure does not allow for a standard investigation plan.

A particularly significant aspect concerns the procedure adopted to verify the guy-wire tension. In this instance, the measurement system served not only a diagnostic purpose but also an operational control function during the test itself. Monitoring the deformation release at the clevis allowed for the detection of the progressive load transfer to the parallel counter-system, preventing the operation from causing an unintended increase in guy-wire tension. In this sense, the diagnostic process also served as a tool for managing the safety of the test.

Dynamic identification and continuous monitoring completed the assessment process by providing a comprehensive overview of the tower's behavior. Natural frequencies,

primary mode shapes, and variations in stay-cable tension—driven by environmental loads—constitute essential data for constructing and calibrating a numerical model that accurately reflects the structure's actual behavior.

The scope of this paper is limited to defining the investigation protocol, conducting the tests, and generating the experimental data required for modeling and subsequent structural assessment. Management decisions, final safety verifications, and the reasons behind the structure's demolition fall outside the study's objectives. However, the fact that the tower no longer exists—having been brought down via controlled collapse in July 2025—gives this study documentary value as well. The investigations conducted serve as a technical record of one of the most significant cable-stayed steel structures built in Italy; while no longer directly observable, it remains a valuable case study for the assessment of unconventional structures.



**Fig. 8** Images of the demolition phases of the RAI tower in Caltanissetta, carried out in July 2025 via controlled collapse.

## 9 Conclusions

This study outlines an integrated investigation protocol applied to the RAI tower in Caltanissetta, selected as a case study of a tall, guyed steel structure characterized by significant knowledge uncertainty. The lack of specific procedural guidelines and the impossibility of conducting extensive destructive testing necessitated an approach combining local measurements, global checks, monitoring, and numerical modeling.

The findings demonstrate that, for data-scarce structures, the reliability of the assessment process depends more on the consistency between independent observables than on the sheer volume of tests performed. Local measurements provide data on materials and stress states; checks on the guy cables allow for the determination of parameters essential to global equilibrium; and dynamic identification and monitoring enable the assessment of the structure's overall behavior under actual operating conditions.

The proposed protocol thus serves as an operational framework applicable to other guyed towers and, more broadly, to unconventional steel structures for which no directly applicable standardized procedures exist. Furthermore, the case study underscores the documentary value of experimental investigations when applied to structures of significant technical and historical importance—particularly those that may not be accessible for future direct inspections.

## Acknowledgment

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